

Eco 202 - Fall 2026

Class 09

$$A = (A \cap B) \cup (A \cap B^c)$$

$$P(A) = P(A \cap B) + P(A \cap B^c)$$

$$= \frac{P(A \cap B)}{P(B)} \cdot P(B) + \frac{P(A \cap B^c)}{P(B^c)} \cdot P(B^c)$$

$$= P(A|B) \cdot P(B) + P(A|B^c) \cdot P(B^c)$$

- $P(A^c|B) = 1 - P(A|B)$

$$1 = P(A^c|B) + P(A|B)$$

$$= \frac{P(A^c \cap B) + P(A \cap B)}{P(B)}$$

$$B = (A \cap B) \cup (A^c \cap B)$$

$$P(B) = P((A \cap B) \cup (A^c \cap B))$$

$$\boxed{=} P(A \cap B) + P(A^c \cap B)$$

by
disjoint

Recall

before : $P(A) = 1 - P(A^c)$

Conditional version

$$P(A|B) = 1 - P(A^c|B)$$

Bayes' rule

$$P[A|B] = \frac{P(A \cap B)}{P(B)}$$

$$= \frac{P(A \cap B)}{P(A)} \cdot \frac{P(A)}{P(B)}$$

$$= P(B|A) \cdot \frac{P(A)}{P(B)}$$